

1. Administrative & Study Identification

Full Study Title: Phase 3 Study of T-DXd and Rilvegostomig Versus SoC in Advanced HER2-expressing Biliary Tract Cancer **Short Title/ Acronym:** DESTINY-BTC01 **Protocol Number:** D781PC00001 **NCT Number:** NCT06467357 **Sponsor Name:** AstraZeneca **Company:** AstraZeneca **Investigational Product:** Trastuzumab Deruxtecan (T-DXd) and Rilvegostomig **Phase of Development:** Phase 3 **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To measure the efficacy and safety of T-DXd combined with rilvegostomig, or T-DXd monotherapy, compared with standard-of-care (gemcitabine plus cisplatin and durvalumab) in patients with advanced, treatment-naïve, HER2-expressing biliary tract cancer (BTC). **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DoR), and safety profiles including the incidence of interstitial lung disease (ILD).

2.2 Background & Rationale Biliary tract cancers are often diagnosed at an advanced stage, rendering systemic therapy the mainstay of treatment. The current first-line standard of care (SoC) consists of gemcitabine plus cisplatin, recently evolved by adding immune checkpoint inhibitors (ICIs) like durvalumab. However, outcomes remain suboptimal. T-DXd is a potent HER2-directed antibody-drug conjugate (ADC) that has shown significant activity in HER2-expressing solid tumors. Rilvegostomig is a bispecific antibody targeting PD-1 and TIGIT. Combining the cytotoxicity of T-DXd with the immune-enhancing effects of rilvegostomig has the potential to overcome the suppressive tumor microenvironment and significantly improve survival outcomes over the current SoC in HER2-expressing BTC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard of Care: gemcitabine + cisplatin [Trade Name: Platinol, ATC Code: L01XA01] + durvalumab) **Blinding:** Open-label **Randomization:** Randomized (Following an initial 20-patient safety run-in)

3.2 Planned Enrollment & Duration Target **N**: 620 participants
Number of Sites: Multicenter, Global (Including US sites like Mayo Clinic, UC San Diego Moores Cancer Center) **Estimated Study Start:** 12-Aug-2024 **Estimated Primary Completion:** 12-Jun-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of informed consent. 2. Unresectable, previously untreated, locally advanced or metastatic biliary tract adenocarcinoma. Prior perioperative/adjuvant treatment is permitted provided there is ≥ 3 months (90 days) between the end of treatment and diagnosis of advanced disease. 3. Histologically confirmed HER2-expressing (IHC 3+ or IHC 2+) BTC. Participants must provide an FFPE tumor sample no older than 3 years for central testing. 4. WHO/ECOG performance status of 0 or 1 with no deterioration over the previous 2 weeks. 5. Minimum life expectancy of 12 weeks. 6. Evidence of post-menopausal status or negative serum pregnancy test for females of childbearing potential.

4.2 Exclusion Criteria 1. History of (non-infectious) Interstitial Lung Disease (ILD)/pneumonitis or current ILD/pneumonitis. 2. Medical history of myocardial infarction within 6 months before randomization, symptomatic congestive heart failure, or recent stroke. 3. Pleural effusion, ascites, or pericardial effusion requiring drainage within 2 weeks prior to screening. 4. Histologically confirmed ampullary carcinoma. 5. Prior exposure to other HER2 targeting therapies, ADCs, immune checkpoint inhibitors, and therapeutic anticancer vaccines. 6. Corrected QT interval (QTcF) prolongation to ≥ 470 msec (females) or ≥ 450 msec (males).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered via cycles. Arm A: T-DXd + Rilvegostomig; Arm B: T-DXd monotherapy. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care, anti-emetics, and hormonal therapy for non-cancer related conditions (e.g., hormone replacement therapy). **Prohibited:** Any concurrent anticancer treatment without an adequate washout period prior to randomization; concurrent uncontrolled infections requiring IV antibiotics, antivirals, or antifungals.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Central HER2/PD-L1 Testing
Baseline	Day 0	Randomization, First Dose, Baseline Performance Status check
Interim	Every Cycle	Safety Labs, Efficacy Assessment, Infusion
End of Study	Upon Progression	Final Vital Signs, Exit Interview, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: - **Safety Run In:** To evaluate the safety and tolerability of T-DXd with rilvegostomig by the proportion of treated patients with AEs, SAEs, and AESIs (CTCAE v5.0). - **Randomized Portion:** Overall Survival (OS) in the Full Analysis Set (FAS) for the HER2 IHC 3+ population. OS is defined as the time from randomization to death due to any cause. **Measurement Method:** Independent Central Review utilizing standard radiological tumor assessments (RECIST v1.1) and survival tracking.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) in the FAS (HER2 IHC 3+/2+) population evaluating T-DXd with rilvegostomig vs SoC. * Overall Survival (OS) in the FAS (HER2 IHC 3+) and (HER2 IHC 3+/2+) populations evaluating T-DXd monotherapy vs SoC. * Objective Response Rate (ORR) * Duration of Response (DoR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring across treatment cycles, with specific protocols for identifying and managing pulmonary symptoms due to the known risk of Interstitial Lung Disease (ILD) associated with T-DXd. **Serious Adverse Events (SAEs):** Reported within 24 hours per standard oncology trial procedures. **Data Monitoring Committee (DMC):** Independent safety review, explicitly required after the initial 20-patient safety run-in phase to clear the study for broader 600-participant randomization.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) / Full Analysis Set (FAS) for primary efficacy outcomes; Safety Analysis Set for all safety and tolerability endpoints. **Sample Size Justification:** \$N = 620\$ randomized participants appropriately powered (\$\alpha = 0.05\$, \$Power = 80\%\$) to detect statistically significant superiority in the experimental arms against the active

comparator (SoC). **Handling of Missing Data:** Standard right-censoring rules applied for time-to-event endpoints (PFS and OS) via Kaplan-Meier methodology.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote clinical monitoring to guarantee protocol adherence. **Audit Trail:** Utilizing validated Electronic Data Capture (EDC) systems with full audit trail capabilities for all eCRFs and central laboratory records.

11. Study Identifiers & Governance

Primary ID: D781PC00001 **Posting ID:** 76655394067198285 **Registry Number:** NCT06467357 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** DESTINY-BTC01 **Official Regulatory Title:** DESTINY-Biliary Tract Cancer-01: A Phase 3 Study of Trastuzumab Deruxtecan (T-DXd) and Rilvegostomig Versus Standard-of-Care Gemcitabine, Cisplatin, and Durvalumab for First Line Locally Advanced or Metastatic HER2-expressing Biliary Tract Cancer

12. Clinical Framework (Biliary Tract Cancer Specific)

Primary Condition / Disease Area: Biliary Tract Cancer **Preferred Name:** Biliary Tract Cancer **Disease Definition:** Biliary tract cancer encompasses a group of malignancies that originate from the epithelial cells lining the bile ducts (cholangiocarcinoma), the gallbladder (gallbladder cancer), or the ampulla of Vater (ampullary carcinoma). These cancers are often aggressive and typically present with subtle symptoms, leading to diagnosis at advanced stages, which can complicate treatment and prognosis.

UMLS Preferred Name: Biliary Tract Neoplasms **Semantic Type:** Neoplastic Process **MeSH Code:** D001661 **SNOMED ID:** 363415003

Diagnosis Threshold: Histologically confirmed HER2-expressing (IHC 3+ or IHC 2+) **Medical Methodology:** Targeted Antibody-Drug Conjugate (ADC) combined with Bispecific Immunotherapy

Optimization Target: Tumor response, halting disease progression, and prolonging overall survival

13. Study Procedures & Methodology

Management Pathway: First-line systemic therapy for advanced BTC

Education Protocol (HCP): Investigator training for early identification and active management of ADC-induced pneumonitis/ILD. **Education Protocol (Patient):** Prompt reporting of respiratory, cardiac, or infusion-related symptoms. **Quality Audit**

